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TECHNICAL NOTE · BATTERY BASICS

Battery % made simple

What SOC and FCC mean, and why the battery percentage sometimes jumps

Think of your battery as a **water tank**.

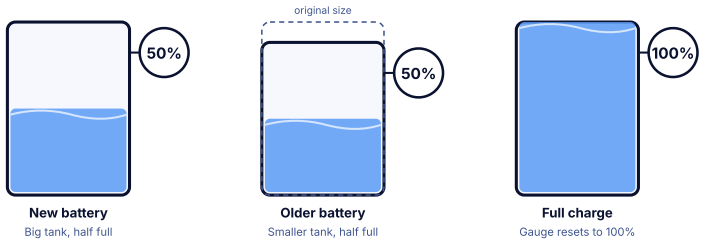
FCC = the size of the tank

How much energy the battery can hold **today**. A new battery is a big tank. As it ages, the tank slowly gets smaller.

SOC = the gauge on the tank

How full the battery is **right now**, in %. This is the number you see in the app.

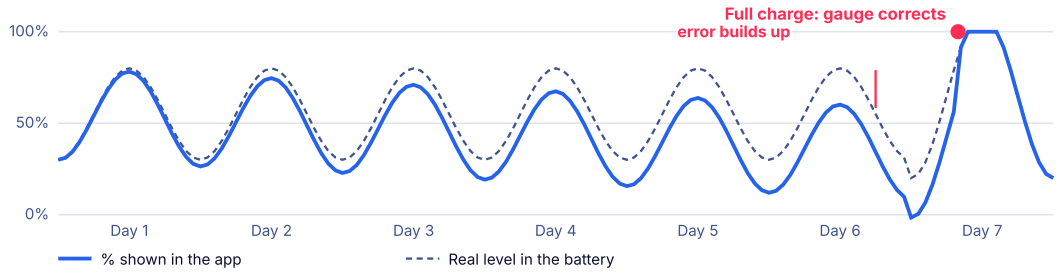
The same 50%, different amounts of energy



Both gauges show 50%, but the older battery holds less energy because its tank (FCC) has shrunk.

Why the % sometimes jumps

The gauge cannot see inside the tank. It **counts** the energy going in and out. Small counting errors add up day after day, so the reading slowly drifts. When the battery charges **completely full**, the gauge knows for sure it is at 100% and corrects itself. That is the sudden jump. The battery is not faulty.



What you should do

1. **Let it charge fully once a week.**
   This lets the gauge reset and keeps the % accurate.
2. **Watch for "full too early".**
   If the battery reaches 100% very early every day, the tank may have shrunk with age. Ask us to check it before buying more battery.
3. **Avoid empty-to-full every day.**
   Deep daily cycles wear the battery faster. Keeping a reserve is kinder to it.
4. **Keep it cool.**
   Install in a shaded, ventilated place, ideally below 40 °C.

In one line: FCC is how much the battery can hold. SOC is how full it is now. The % is an estimate, and it needs a full charge now and then to stay accurate.